

ADVICE FROM LAST YEAR FINALIST - Super important to systematically lay out what the problem is (what last mile is, what the issue with it is, lack of communication), then define how Amazon can solve this problem and why Amazon is the BEST company to solve this problem (compared to competitors like Tesla/Space X). Lowk PROBLEM/SOLUTION are super important to determine the trajectory of the presentation.

New info after some research - we should only focus on implementing this strategy in the **US** in the short term (next 5-10 years). Global expansion is NOT realistic due to geographical blind spots, legislative limitations, etc. Amazon operates in A LOT of countries, so we risk having too large of a scope if we try to include them all. We can *maybe* mention Canada/Mexico if we want to discuss long-term possibilities.

Problem - JR

- Last mile delivery
 - Easily gets to closest warehouse
 - Especially in rural areas, takes days to deliver to individual houses
 - WHY IS THIS A PROBLEM
 - Small scattered deliveries -- high cost, low efficiency
 - Driver scheduling (set routes, limited capacity)
 - Lack of drivers
 - Rural areas
 - Even more costly - no network that GPS relies on
 - Spotty GPS (hard to have live updates/find most efficient path)
 - Hard to locate car AND whether package is ready/delivered to nearest warehouse already

Solution - Esther

- Utilize Zoox (driverless taxis) to deliver from warehouses to houses
- Adding mobile locker compartment in zoox
 - Scan QR code to open locker
 - Prevent theft
 - Student mail room vibe
- Schedule packages to come to houses when people are home (can pick it up off of zoox)
 - Can provide
- Look at scheduling to maximize most efficient routes using LEO satellite GPS
 - Gives better real-time updates (new traffic, upsets, etc)
 - In rural areas, allow customers to see real-time updates
 - Utilize fast LEO satellites

WHAT QKUIPER DOES - A BRIEF OVERVIEW:

- A network of 3,000+ low Earth orbit satellites providing global internet

- Designed to deliver fast, low-latency connectivity anywhere, including remote areas
- Kuiper does not directly deliver packages, but instead addresses the infrastructure constraint behind last-mile delivery
- Traditional delivery struggles in areas with weak internet and network coverage
- Kuiper enables reliable connectivity even in remote or underserved regions
- This allows Amazon to serve areas that were previously too costly or inefficient to reach
- Expands Amazon's total addressable market by improving geographic coverage
- Last-mile delivery relies heavily on real-time data such as traffic, weather, delivery density, and customer availability
- Kuiper enables continuous connectivity across drivers, vehicles, and logistics systems
- This improves route optimization, reduces failed deliveries, and increases delivery density per route
- Kuiper enables connectivity across the entire delivery ecosystem including vans, drones, robots, and warehouses
- Supports real-time communication across all nodes in the network
- Enables future capabilities such as autonomous delivery, smarter dispatching, and faster issue resolution
- Kuiper integrates with AWS to create a seamless data pipeline from delivery operations to cloud-based optimization systems
- Data can flow from delivery → satellite → cloud → optimization system in near real time
- This improves decision-making speed and overall operational efficiency
- Current last-mile systems depend heavily on third-party cellular networks and local telecom infrastructure
- Kuiper enables a more vertically integrated system controlled by Amazon from space to cloud to delivery
- Reduces reliance on external providers and increases control over cost, reliability, and scalability



1. Build Autonomous Last-Mile Fleet (Zoox + Smart Lockers)

What

- Deploy Zoox vehicles equipped with **secure mobile locker compartments**
- Customers unlock via QR code → eliminates theft + missed deliveries

Why it solves the problem

- Reduces dependency on human drivers → addresses labor shortages
- Converts **door-to-door delivery into scheduled pickup points on wheels**
- Increases delivery density (one vehicle → multiple households)

How it scales

- Start in:
 - suburban + semi-rural test markets
- Expand as autonomy improves + regulatory approval increases

2. Kuiper-Enabled Real-Time Routing & Dispatch Optimization

What

- Use Amazon Project Kuiper to power:
 - real-time GPS + routing
 - dynamic rerouting based on traffic, weather, and delivery readiness

Why it solves the problem

- Fixes **spotty GPS + connectivity gaps (especially rural)**
- Enables:
 - live vehicle tracking
 - smarter route clustering
- Reduces idle time (e.g., arriving before package is ready)

How it scales

- Integrate Kuiper → AWS → logistics systems
- Continuous improvement via data feedback loops

3. Demand-Synchronized Delivery Scheduling (Customer + System Alignment)

What

- Shift from “deliver ASAP” → **deliver when customer is available**
- Customers select:
 - time windows
 - or receive optimized suggestions

Why it solves the problem

- Eliminates:
 - failed deliveries
 - re-delivery costs
- Increases **drop success rate per trip**
- Matches:
 - vehicle routes ↔ customer presence

How it scales

- Use predictive models (via AWS) to:
 - cluster deliveries geographically + temporally

- Incentivize users:
 - discounts for flexible time slots

4. Create a Fully Connected Logistics Ecosystem (Kuiper + AWS + IoT)

What

- Connect all nodes:
 - warehouses
 - Zoox vehicles
 - packages
 - customers
- Through:
 - Kuiper (connectivity)
 - Amazon Web Services (data processing)

Why it solves the problem

- Eliminates **information fragmentation**
- Enables:
 - real-time package status
 - coordination between warehouse + vehicle
- Reduces:
 - uncertainty (Is package ready? Where is vehicle?)

How it scales

- Builds Amazon-owned, vertically integrated infrastructure:
 - space → cloud → delivery
- Reduces reliance on third-party telecom networks

Financials - Amy

Conclusion - All